

Disproof of conjecture 00000002853: cactus rank already falls below Waring rank in degree 3

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1 The conjecture and the reading

Statement (English, verbatim). Definition: Cactus rank and algebraic statistics: the statistical translation of apolarity theory. Conjecture: The stratification rigidity of cactus rank: the minimal degree where cactus rank falls below rank is 5, and the algebraic mechanism of the separation is the multi-point splitting constraint of apolar ideals. (cactus separation minimal degree)

The Chinese version makes the same claim: the minimal degree at which the cactus rank is lower than the rank is 5, and the mechanism of the separation is a multi-point splitting constraint of apolar ideals.

Reading. “Rank” is the Waring rank (the rank notion of apolarity theory). The first conjunct says: if a nonzero complex form F of degree d has cactus rank strictly smaller than its Waring rank, then $d \geq 5$, and degree 5 occurs. We refute this conjunct, which refutes the conjunction. The second conjunct (“the mechanism is the multi-point splitting constraint”) has no precise meaning and is not addressed. We work over $\mathbb{C}$, in any finite number of variables; the counterexample is a binary cubic.

2 Definitions (as formalized)

Let $S = K[x_0, \dots, x_{n-1}]$ (forms) and let $T = K[y_0, \dots, y_{n-1}]$ be the dual ring. In Lean both are `MvPolynomial (Fin n) K`.

Definition 1 (Apolarity). T acts on S by differentiation: $y^m \circ F = \partial_0^{m_0} \dots \partial_{n-1}^{m_{n-1}} F$ with $\partial_i = \partial / \partial x_i$, extended linearly: $g \circ F = \sum_m g_m \partial^m F$ (`partialPow`, `diffAct`). The apolar set is $F^\perp = \{g \in T : g \circ F = 0\}$ (`annDiff`). As a second convention we also use contraction, $y^m \lrcorner x^a = x^{a-m}$ if $m \leq a$ and 0 otherwise (`contractAct`, `annContract`).

Bernardi and Ranestad [BR] use contraction and remark: “Differentiation may be used instead of contraction, if care is made with coefficients.” For our monomial F both conventions give the same result, and the Lean file proves the result for both.

Definition 2 (Zero-dimensional subscheme of length r). An ideal $I \subseteq T$ is *homogeneous* if it contains every homogeneous component of each of its elements (`IsHomogeneousIdeal`). With $\mathfrak{m} = (y_0, \dots, y_{n-1})$ (`irrelevant`), I is *saturated* if $(I : \mathfrak{m}^\infty) = \bigcup_k (I : \mathfrak{m}^k) \subseteq I$, i.e. $hg \in I$ for all $h \in \mathfrak{m}^k$ implies $g \in I$ (`IsSaturated`). The Hilbert function is $t \mapsto \dim_K(T/I)_t$ (`hilbertFunction`). I is the ideal of a zero-dimensional subscheme of length r (`IsZeroDimLength I r`) if it is homogeneous, saturated, and its Hilbert function equals r for all large t .

This encodes the standard dictionary: closed subschemes of $\mathbb{P}^{n-1} = \text{Proj } T$ correspond bijectively to saturated homogeneous ideals (Wikipedia, *Projective variety*: “correspond bijectively to the homogeneous ideals”), and the dimension and degree are read off the Hilbert polynomial (ibid.: “Basic invariants of X such as the degree and the dimension can be read off the Hilbert polynomial of this graded ring.”). A subscheme is zero-dimensional of length r exactly when its Hilbert polynomial is the constant r , that is, when the Hilbert function of T/I_Z equals r in all large degrees. Saturation is the ideal quotient of Wikipedia, *Ideal quotient* (“the saturation of an ideal corresponding to a projective scheme”).

Definition 3 (Cactus rank and Waring rank). A subscheme with ideal I is *apolar* to F if $I \subseteq F^\perp$ ([BR], Definition 1: “A subscheme is apolar to if its homogeneous ideal is contained in”). The *cactus rank* $\text{cr}(F) \in \mathbb{N} \cup \{\infty\}$ is the least r such that some I with $\text{IsZeroDimLength } I \ r$ satisfies $I \subseteq F^\perp$ (`cactusRank`; `cactusRankContract` for contraction). [BR]: “The cactus rank is the minimal length of an apolar subscheme to”; [BT]: “The cactus rank of a homogeneous polynomial is known to be the minimal length of an apolar zero-dimensional scheme to”. The *Waring rank* $\text{r}(F)$ of a form of degree d is the least r with $F = \sum_{i=1}^r L_i^d$, where $L_i = \sum_j a_{ij} x_j$ are linear forms (`waringRank d F`); Wikipedia, *Comon’s conjecture*: “the minimal number of d -th powers of linear forms needed to express it”. Both infima are taken in $\mathbb{N} \cup \{\infty\}$ (`ENat`), with value ∞ if no witness exists.

The set of separating degrees is $\mathcal{D} = \{d > 0 : \exists n, \exists F \in \mathbb{C}[x_0, \dots, x_{n-1}] \setminus \{0\} \text{ homogeneous of degree } d \text{ with } \text{cr}(F) < \text{r}(F)\}$ (`separatingDegrees`). Degrees are required to be positive, because the coefficient-free Waring-rank definition $F = \sum L_i^d$ is standard only for $d > 0$; at $d = 0$ it would give a nonzero constant c the meaningless value $\text{r}(c) = c$. The conjecture’s first conjunct says that 5 is the least element of $\mathcal{D}$.

3 The counterexample $F = x_0^2 x_1$

Lemma 4 (The double point). $I_0 = (y_1^2) \subseteq \mathbb{C}[y_0, y_1]$ is homogeneous and saturated, and $\dim(T/I_0)_t = 2$ for every $t \geq 1$. So I_0 is the ideal of a zero-dimensional scheme of length 2 (the double point at $[1 : 0]$).

Proof. I_0 is the monomial ideal generated by y_1^2 , so $g \in I_0$ iff every monomial y^m of g has $m_1 \geq 2$. Homogeneous components keep a subset of the monomials, so I_0 is homogeneous. If $hg \in I_0$ for all $h \in \mathfrak{m}^k$, take $h = y_0^k$. The monomials of $y_0^k g$ are $y_0^k y^m$ for y^m in g , and these have the same y_1 -exponent, so $g \in I_0$. By the standard-monomial count (dimension of $(T/I)_t =$ number of degree- t monomials not divisible by any generator), $(T/I_0)_t$ has basis $y_0^t, y_0^{t-1} y_1$ for $t \geq 1$. $\square$

Lemma 5 (Apolarity). $I_0 \subseteq F^\perp$ for $F = x_0^2 x_1$, for differentiation and for contraction.

Proof. Each monomial y^m of $g \in I_0$ has $m_1 \geq 2$. Then $\partial^m F = \partial_0^{m_0} \partial_1^{m_1-2} (\partial_1^2 F) = 0$ because $\partial_1^2 (x_0^2 x_1) = 0$. For contraction, $m \leq (2, 1)$ fails because $m_1 \geq 2 > 1$. $\square$

Hence $\text{cr}(F) \leq 2$ (`cactusRank_F0_le`, `cactusRankContract_F0_le`).

Lemma 6 (Waring rank). $\text{r}(x_0^2 x_1) = 3$.

Proof. Upper bound: $6x^2y = (x+y)^3 - (x-y)^3 - 2y^3$. With real $c = 6^{-1/3}$ and $e = 2^{1/3}$ this gives $x^2y = (cx+cy)^3 + (-cx+cy)^3 + (-ecy)^3$. Lower bound: suppose $x^2y = \sum_{i \leq r} (\alpha_i x + \beta_i y)^3$ with $r \leq 2$. Pad with zero forms to $r = 2$ and put $x = 1, y = t$, so that $t = f(t) := (\alpha_1 + \beta_1 t)^3 + (\alpha_2 + \beta_2 t)^3$ for

all t . Evaluating at $t = 0, 1, -1, 2$ and solving the linear system gives $\beta_1^3 + \beta_2^3 = 0$, $\alpha_1\beta_1^2 + \alpha_2\beta_2^2 = 0$ and $3(\alpha_1^2\beta_1 + \alpha_2^2\beta_2) = 1$. The identity

$$\beta_1^3 = -\beta_1^3(3(\alpha_1^2\beta_1 + \alpha_2^2\beta_2) - 1) + 3(\alpha_1\beta_1^2 - \alpha_2\beta_2^2)(\alpha_1\beta_1^2 + \alpha_2\beta_2^2) + 3\alpha_2^2\beta_2(\beta_1^3 + \beta_2^3)$$

gives $\beta_1^3 = 0$, so $\beta_1 = 0$, then $\beta_2 = 0$, and $3(\alpha_1^2\beta_1 + \alpha_2^2\beta_2) = 0 \neq 1$. $\square$

Theorem 7. $\text{cr}(x_0^2x_1) \leq 2 < 3 = \text{r}(x_0^2x_1)$, so $3 \in \mathcal{D}$. Hence 5 is not the least element of $\mathcal{D}$, and $\min \mathcal{D} \leq 3$. The first conjunct of the conjecture, and with it the conjecture, is false.

Remarks (not formalized). Degrees 1 and 2 do not separate: for quadrics the Waring rank, the cactus rank and the matrix rank all coincide. So the true minimal degree is 3. Other readings also point to degree 3. [BR] shows “there are cubic forms with minimal cactus rank (equal to the Hankel rank) whose border rank is strictly higher”, and its abstract states that, for a general cubic form in sufficiently many variables, the smallest length of an apolar zero-dimensional scheme is bounded “and therefore smaller than the rank of the form”. These are cited only as context, and the Lean proof does not use them.

4 Lean formalization

File `Conjecture2853/Basic.lean` (namespace `C2853`), import `Mathlib` only. The monomial-ideal Hilbert-function machinery (`monomialIdeal`, `gradedPiece`, `hilbertFunction`, `standard`, `hilbertFunction_monomialIdeal`) is copied, with credit, from the accepted solution of conjecture 00000000539. Main results:

- `I0_zeroDim : IsZeroDimLength (I0 K) 2; I0_apolar, I0_apolar_contract;`
- `waringRank_F0: waringRank 3 (F0 C) = 3 (from not_two_cubes);`
- `F0_separates: cactusRank (F0 C) < waringRank 3 (F0 C) and cactusRankContract (F0 C) < waringRank 3 (F0 C);`
- `three_mem_separatingDegrees: 3 ∈ separatingDegrees;`
- `conjecture_false: 3 ∈ separatingDegrees ∧ ¬IsLeast separatingDegrees 5 ∧ sInf separatingDegrees ≤ 3`, where `separatingDegrees` contains only positive degrees ($0 < d$ is part of its definition).

`lake build` succeeds. `#print axioms` for `conjecture_false`, `F0_separates` and `waringRank_F0` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.

References

- [BR] A. Bernardi, K. Ranestad, *On the cactus rank of cubic forms*, <https://arxiv.org/abs/1110.2197>.
- [BT] A. Bernardi, D. Taufer, *Waring, tangential and cactus decompositions*, <https://arxiv.org/abs/1812.02612>.
- Wikipedia: https://en.wikipedia.org/wiki/Projective_variety, https://en.wikipedia.org/wiki/Ideal_quotient, https://en.wikipedia.org/wiki/Comon%27s_conjecture.